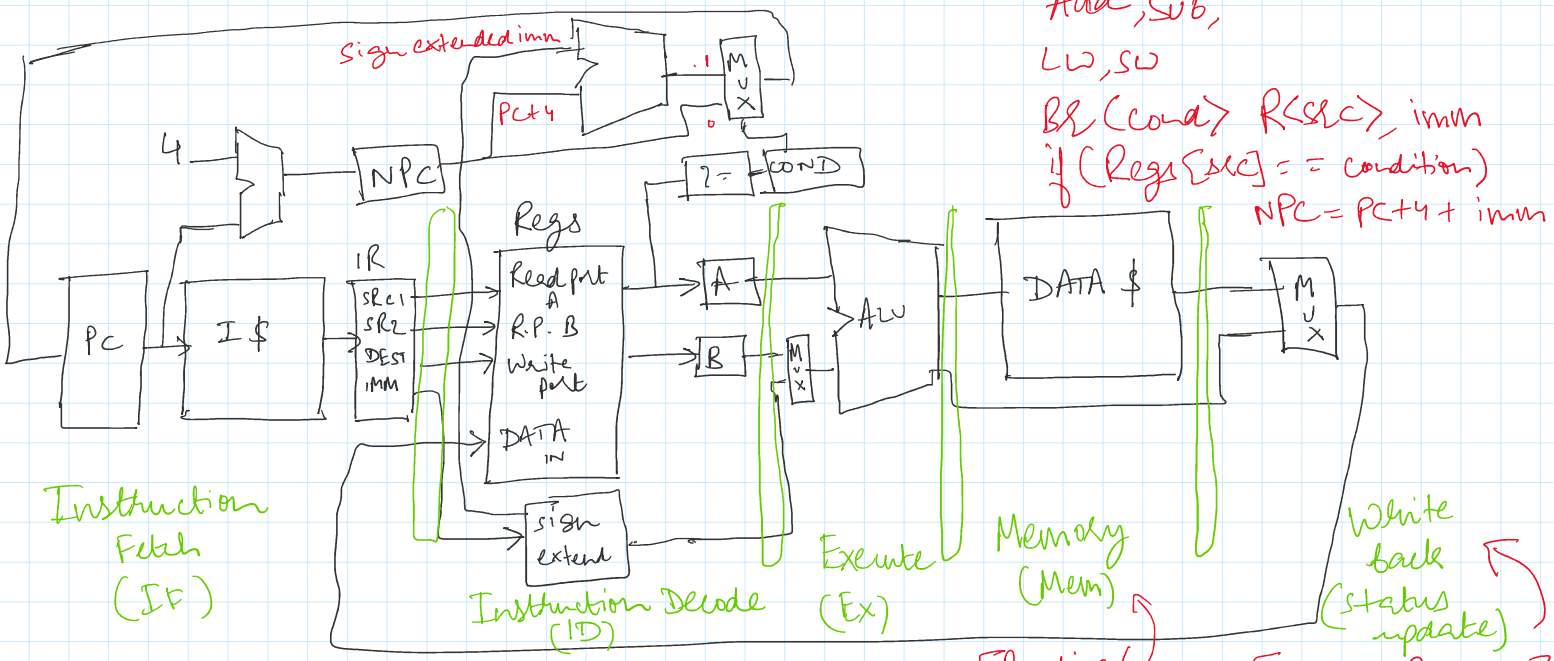
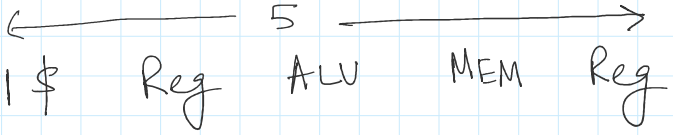
Project 1 - Due tomorrow
Homework 3 - pipelining (most likely out tomorrow)

Last Time
Hit time reduction in $ → Interaction with memory → VIPT $ → virtualization
Tiny ISA description - 3 types of instructions

Today - Build Hardware for tiny ISA + pipline it



Add, Sub,
LW, SW
BR (cond) R<src>, imm
if (Regs[src] == condition)
NPC = PC+4 + imm



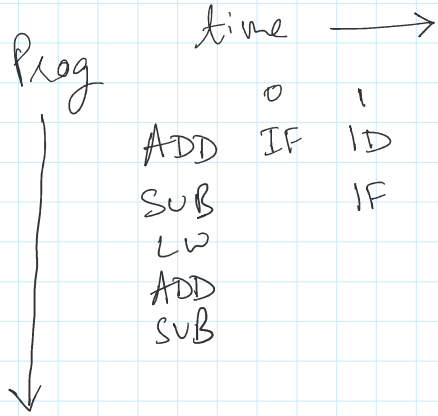
I $ Reg ALU MEM ALU

CPI = 5

	0	1	2	3	4	5	6
ADD	I$	Reg	ALU	MEM	Reg		
LW		I$	Reg	ALU	Mem	Regs	
SUB			I$	Regs	ALU	MEM	Reg

Adding latches to isolate each stage of the datapath

Waterfall / Pipeline Diagram



$$\text{Speedup} = \frac{\text{exec. time unpipelined}}{\text{exec. time pipelined}} = \frac{IC \times CPI_{unpipe} \times CT}{IC \times CPI_{pipe} \times CT}$$

$$= \frac{CPI_{unpipelined}}{\frac{CPI_{unpipelined}}{5} + \text{Latching overhead}}$$

$$= 5$$

N stage pipeline, speedup = N

Pipeline Hazards: May lead to a pipeline stall

1. Data (Flow) hazard – Dependencies b/w instructions preventing overlapped execution
2. Control (Flow) hazard – A branch may change PC but not until later
3. Structural hazards – Not enough hardware resources for combination of instructions

Example of structural hazard: Unified I/D $

Accesses memory

	0	1	2	3	4	5	6	7	8	9
LW	IF	ID	EX	MEM	WB					
SUB		IF	ID	EX	MEM	WB				
ADD			IF	ID	EX	MEM	WB			
NOT				stall	IF	ID	EX	MEM	WB	
OR						IF	ID	EX	MEM	WB